

# How it works (plain-language guide)

This document explains what the Library Contents Claim Agent does, step by step, without code. It is written for an adjuster, a product owner or anyone who wants to understand the flow before reading the technical guide (<docs/pdf/technical-guide.pdf>).

## The problem it solves

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After a fire, flood or burglary, a policyholder has to list every book and object in a room, say what each one is worth, and prove it. Today that means a spreadsheet, photos, and a site visit. This agent turns that into **one walk around the room with a phone**, talking to an assistant while the inventory writes itself. The output is a claim file an adjuster can check line by line.

## The one rule behind everything

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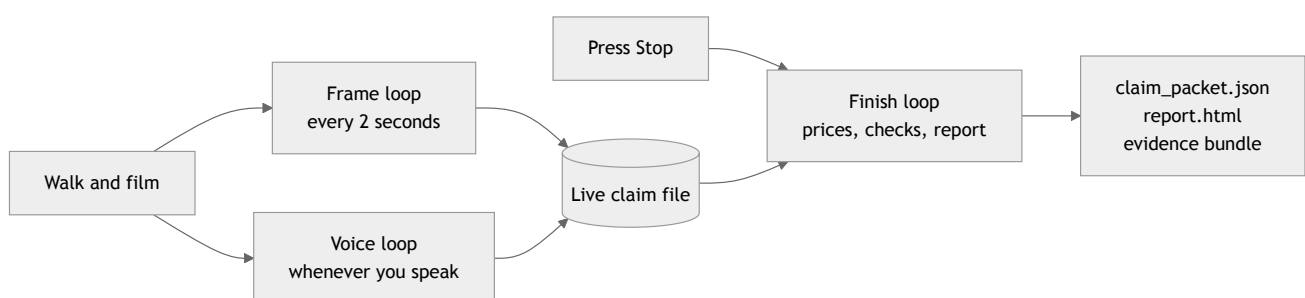
**Software may propose; only evidence may decide.**

- A camera model may say "that looks like a book". It cannot decide the book exists.
- A text model may say "the title seems to be Dune". It cannot decide the title.
- A price only exists when it comes from a real listing with a link and a date.
- Every total is added up in ordinary code, never by an AI.

If any step is unsure, the line is left blank and put in a review list with the reason. A blank is always preferred over a guess, because a guess cannot be defended in a dispute.

## The three loops

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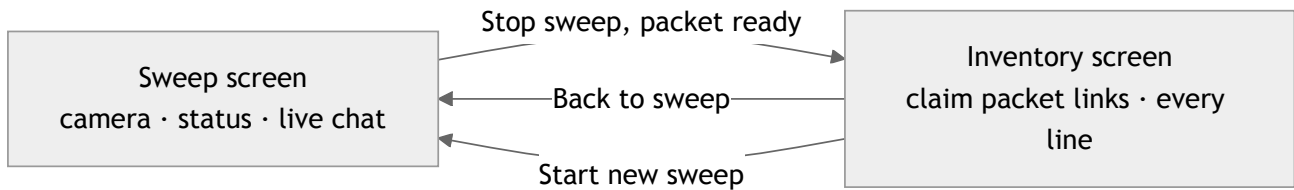
The whole program is three loops working on one shared claim file. Two run while the camera is on; the third runs once when you stop.

## Step by step: starting a sweep

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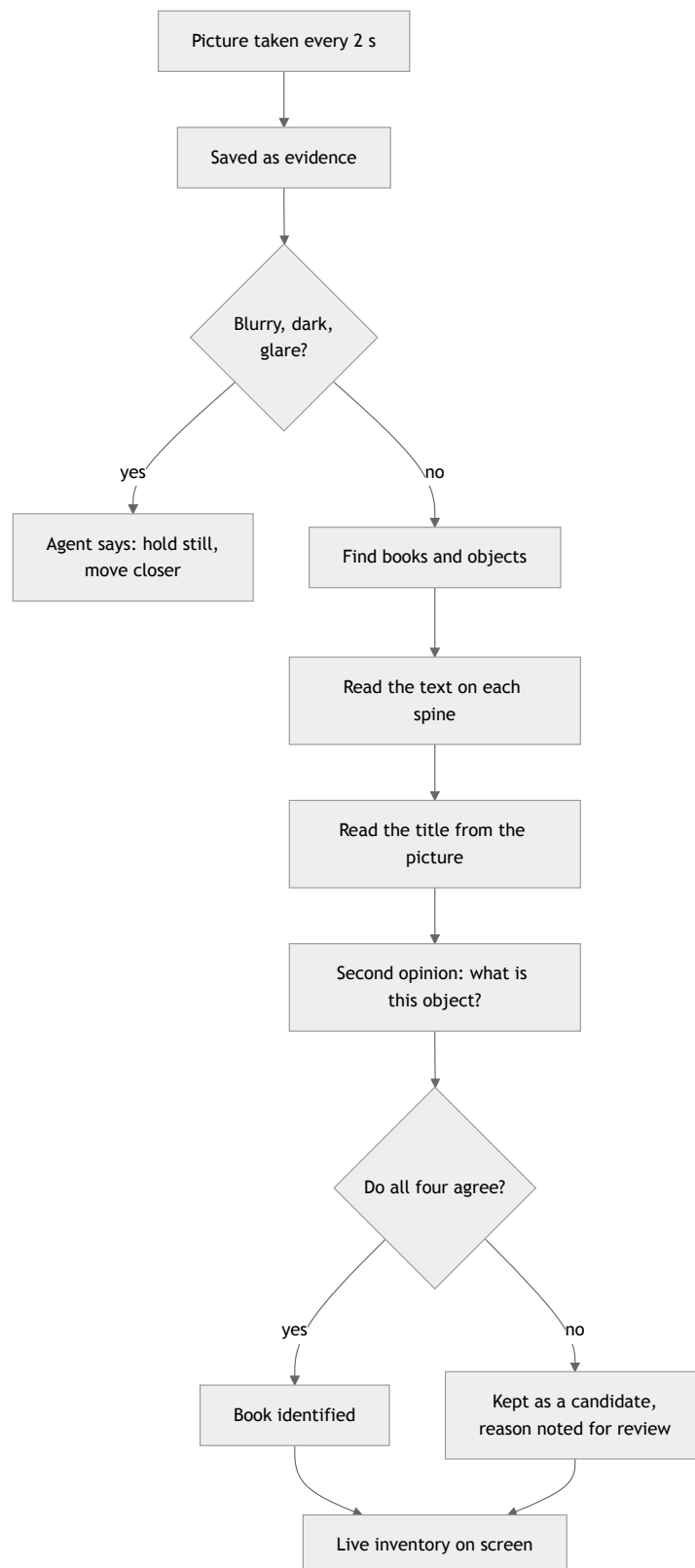
1. Open the page, pick the **country**. The currency follows automatically. This matters because the same book has a different replacement price in London and in Dubai.

2. Set the **appraisal threshold** (default 2,000 in the claim currency). Anything worth more than this is never priced automatically; it is sent to a human appraiser.
3. Press **Start sweep**. The browser asks for the camera and microphone. The agent greets you, confirms the country and currency, and explains what to do in two sentences.
4. **Two screens**. The sweep screen has the camera on the left and the live chat with the agent on the right, with a coloured status line under the camera (green: scanning; amber: processing a picture, with the seconds counting; blue: finishing) and a small line of counts. The inventory screen opens from the **Inventory & packet** button once the sweep is finished, and the browser's back button returns to the sweep.



## Step by step: the frame loop (what happens to every picture)

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1. **Picture saved.** Every frame is stored on disk with a name. Every line in the final claim points back to the frame it came from, so an adjuster can look at the actual photo.
2. **Quality check.** Blur, darkness and glare are measured. If the picture is bad, the agent tells you how to fix it instead of trying to read it. It also looks at the boxes it found: when a third of the spines are cut off at the edge it says "step back a little", and when the spines are small it says "move closer, one shelf at a time".

3. **Find the books and objects.** Three detectors draw boxes: two general ones and one that is told the vocabulary of a home library (shelving, lamp, framed painting, rug, book spine). Everything that is not a book (television, chairs, vases, a lamp) becomes an "item".
4. **Read the spine text.** Small spines are enlarged and their contrast stretched, then the text is read by OCR upright, rotated both ways and, if nothing was readable, upside down, because old spines are printed in all directions.
5. **Read the title from the picture.** A small vision model looks at the same crop and says what the main title is. It may only copy an author or publisher that the OCR also saw, and a person's name is never accepted as the title. The reader gets 60 seconds per picture, best-readable spines first; the rest are read after you press Stop, so a dense shelf cannot freeze the sweep.
6. **Second opinion.** A different vision model is shown the crop with no hint, asked "what is this?" and asked to write down the words it can see. If it does not say "book" (for example "tissue box"), the line cannot be identified and the disagreement is written down. The same check tells a mirror from a framed print, and the agent then asks "original or print?".
7. **Agreement.** A book is marked **identified** in one of two ways, both bound to what is on the spine: the proposed title is literally in the OCR text, or the two vision models read the same title independently and the OCR fragments match at least a third of its letters in order. In both cases a second detector must have drawn the same box and the spine must be whole. If OCR read a word cleanly but the second model read it differently ("MASTERPIEG" against "masterpiece"), the title is refused as a probable misread.
8. **Several pictures of one spine.** The same book seen in more than one picture keeps all of its evidence together: a spine cut off in one picture but whole in the next counts as whole, and the agreement check is run again on the pooled evidence.
9. **Live counts.** The sweep screen updates after every frame: how many books seen, how many identified, how many items, how many need review. The full list is on the inventory screen when the sweep is finished.

## Step by step: the voice loop (talking to the agent)

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The agent listens the whole time. There are two kinds of sentences.

**Commands** change the claim and are matched exactly, so a stray word cannot trigger them:

| You say                                   | What happens                                                                   |
|-------------------------------------------|--------------------------------------------------------------------------------|
| "next shelf"                              | the next frames are labelled as the next shelf                                 |
| "skip this shelf, those are not mine"     | that shelf's books are removed from the claim but kept in the evidence history |
| "that is a first edition" / "signed copy" | the selected book is flagged for human appraisal                               |
| "it is a print" / "it is an original"     | answers the agent's question about a picture on the wall                       |
| "the shelf is 90 centimetres wide"        | gives the real-world scale, so book heights and thicknesses are measured in cm |

| You say                                      | What happens                                               |
|----------------------------------------------|------------------------------------------------------------|
| "the room is 4.2 by 3.1 metres, ceiling 2.6" | records the room size; floor and wall areas are calculated |
| "compare prices in the United Kingdom"       | re-prices ten books for a second country                   |
| "all shelves captured"                       | records that you confirm the room is fully covered         |

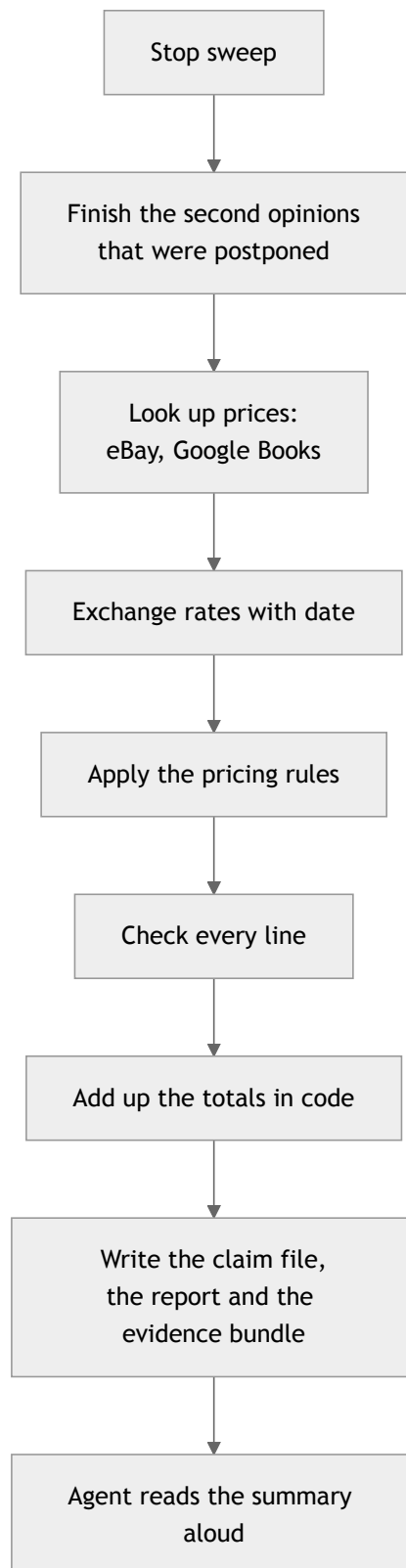
**Questions** ("how many books so far?", "what should I do next?") are answered by a small local chat model that can see the live inventory. It can describe the claim but it is not allowed to change it.

If you talk while the agent is speaking, the agent stops; the conversation is two-way. You can also type the same sentences in the chat box on the right of the sweep screen.

## Step by step: finishing (what happens when you press Stop)

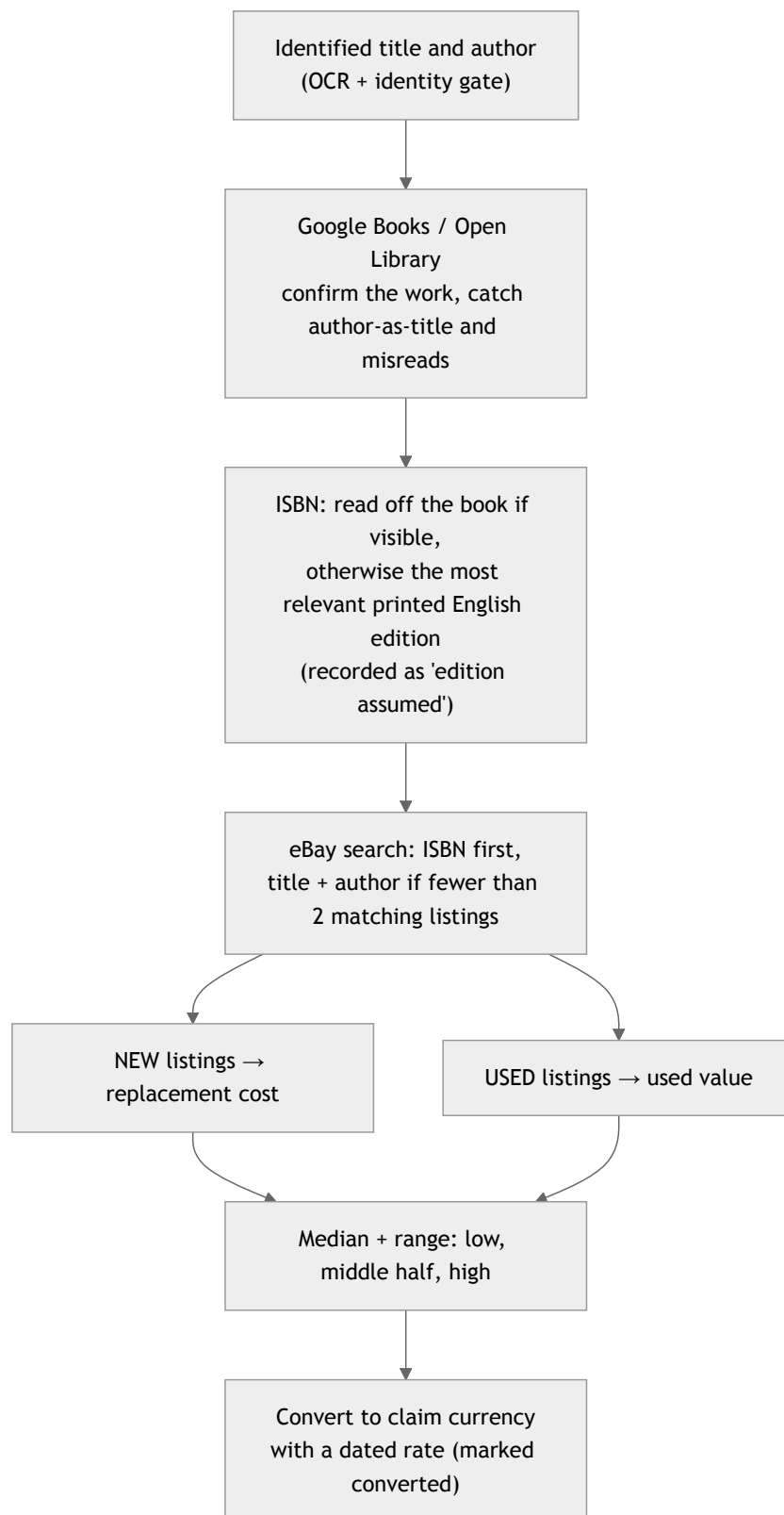
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If a frame is still being analysed when you press Stop, the screen says "Finishing: waiting for the last frame" and waits for it. Nothing is lost and nothing needs to be pressed again.



1. **Postponed work.** During the sweep each picture gets 60 seconds of title reading and 90 seconds of second opinions; the rest are done now, and the status line shows "After- capture checks 4 of 47". Every accepted title is then checked against the Open Library catalogue: a title that is really an author's name is moved to the author field, and a title one wrong letter away from a known book is flagged as a probable misread. Both are kept for review, not priced.
2. **Price lookup.** For every identified book:
  - the title and author are checked in the Google Books and Open Library catalogues;

- an ISBN is taken from the book itself if one was visible, otherwise from the catalogue (the most relevant printed English edition), and marked "edition assumed" because one title can have hundreds of editions;
- eBay is searched by that ISBN, and by title and author if the ISBN finds fewer than two matching listings;
- new listings give the replacement cost and used listings give the used value: the middle listing is the price, and the report also shows how many listings there were, the lowest, the highest and the middle half. Each candidate keeps its link, date, condition and currency. These are current asking prices; eBay's sold prices are not publicly available.



3. **Exchange rates.** If a price is in another currency, a dated central-bank rate converts it and the line is marked "converted" with the rate and its source.
4. **Pricing rules.** A price typed by an operator beats a local listing, which beats a converted foreign one. Weak matches and e-books are rejected with a reason. Signed, rare and first editions, original art and anything above the threshold go to appraisal.

5. **Checks.** Every line is tested: is the title readable, is there a price with a source and a date, does the currency match, is the measurement plausible, is the evidence frame on disk. Failures go to the review list.
6. **Totals.** Added in code from the lines that have values. Lines without a value are counted as "excluded from totals" so nobody mistakes a blank for a zero.
7. **Deliverables.** `claim_packet.json` in the agreed structure, `report.html` for reading, and a zip with both plus every referenced photo and a checksum list.
8. **Summary.** The agent reads back the counts and totals, generated from the numbers.

## What the adjuster receives

| File                           | Purpose                                                                                                       |
|--------------------------------|---------------------------------------------------------------------------------------------------------------|
| <code>claim_packet.json</code> | the structured claim: sweep, room, books, items, totals, review queue, then the evidence behind each decision |
| <code>report.html</code>       | the same content as a readable report with photo links and the price comparison table                         |
| <code>bundle.zip</code>        | packet, report, all referenced frames and a SHA-256 manifest to prove nothing changed                         |

## What it costs and how long it takes

- **Model cost: zero.** All AI runs on the laptop. The only outside calls are free price and exchange-rate lookups, and their count is recorded in the packet.
- **Time:** an empty view takes about 3 seconds. A shelf picture is capped at roughly four minutes on an 8 GB laptop (detection, OCR up to 60 s, title reading 60 s, second opinions 90 s); the rest of the reads happen after Stop. Measured on a worn 53-spine shelf: 274 s for the picture, 11 s to the first packet, then the postponed checks in the background.

## Honest limits

- Without eBay credentials, most books will have no price. The packet says so rather than inventing one. Google Books without an API key runs into a shared daily quota.
- Dense shelves of very small, worn spines are detected but rarely identified. On the test shelf most spines failed because the photo cut them off at the edge or only one detector drew the box; a real sweep with margins around the shelf avoids both. Old spines with the author in large type and the title in small type stay in the review list, which is the right outcome for a claim.
- Measurements need you to state one real dimension; nothing is measured from pixels alone.
- The 60-book ground-truth test, the demo video and the measured results sheet still have to be recorded in a real room.